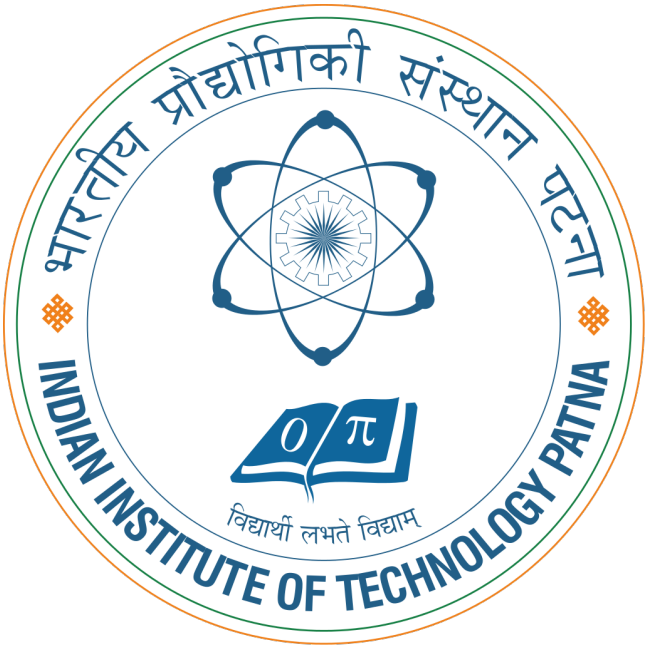
Spoken Digit Sequential Classification (Pre-computed MFCC Sequences)



Ritik Yadav (2621MA03)
Sahnil Siddique (2621MA02)
Suman Acharya (2621MA11)

Department of Mathematics
Indian Institute of Technology Patna

1 Introduction

This task classifies spoken Arabic digit utterances using pre-extracted Mel-Frequency Cepstral Coefficients (MFCCs). Each utterance is a variable-length sequence of 13-dimensional feature vectors. Recurrent models learn temporal patterns from these sequences and assign each utterance to one of ten digit classes (0–9).

2 Dataset and preprocessing

The notebook reports 6,600 training utterances and 2,200 official test utterances. Each line in an utterance block contains 13 MFCC values. Sequence lengths range from 4 to 93 frames, with a mean length of 39.8 frames. Training utterances are split into 6,000 training examples and 600 validation examples; the official test set contains 2,200 utterances.

MFCC coefficients are standardized using mean and standard deviation calculated from the training split only. Variable-length sequences are padded within batches, while the true lengths are retained so padded frames do not influence the final prediction. The validation split is speaker-independent according to the notebook description.

3 Sequence-length distribution



Figure 1: Distribution of utterance lengths in frames.

4 MFCC sequence visualization

An MFCC heatmap displays feature coefficient index vertically and time frames horizontally. Differences in the heatmaps illustrate how acoustic patterns vary across utterances and spoken digit classes.

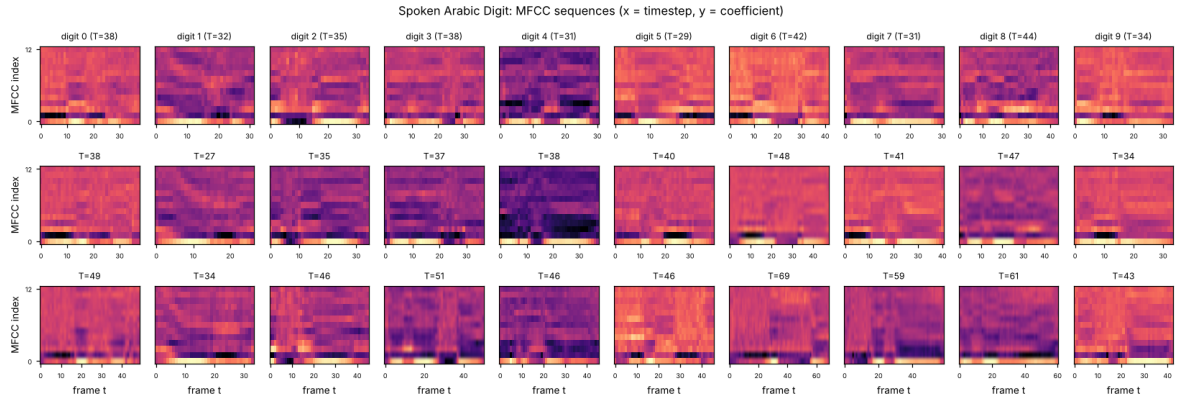


Figure 2: MFCC heatmaps for three training speakers across all ten digit classes.

5 Model architectures and training

Vanilla RNN: updates a hidden state at every time frame. It is a simple baseline but can have difficulty preserving long-range information.

LSTM: uses input, forget and output gates with a memory cell to retain useful information across long sequences.

GRU: uses update and reset gates and often offers a smaller gated architecture than LSTM.

Bidirectional models: process the sequence in forward and reverse directions. Temporal attention pooling learns to weight informative frames when forming an utterance-level representation.

The notebook uses a convergence criterion based on the absolute difference between average training errors in successive epochs falling below the specified threshold, rather than relying solely on a fixed epoch count.

6 Hyperparameter experiment results

Configuration	Epochs	Train (%)	Val (%)	Test (%)
RNN-L1-H64	40	82.55	83.83	76.41
RNN-L1-H128	45	95.72	95.67	88.86
RNN-L2-H64	44	94.58	91.00	85.59
RNN-L2-H128	39	97.33	92.83	90.18
LSTM-L1-H64	37	99.28	97.50	92.86
LSTM-L1-H128	29	99.88	98.00	94.27
LSTM-L2-H64	29	99.92	98.17	94.64
LSTM-L2-H128	22	99.95	98.00	94.50
GRU-L1-H64	38	98.98	99.00	94.05
GRU-L1-H128	34	99.75	98.83	94.82
GRU-L2-H64	31	99.78	98.50	93.86
GRU-L2-H128	27	99.95	98.83	95.27
BiLSTM-L1-H128	20	99.98	95.83	96.95
BiGRU-L1-H128	23	99.97	98.50	96.73
LSTM-L1-H128 + attention	16	99.47	96.33	95.32
GRU-L1-H128 + attention	25	99.68	97.33	96.68
BiGRU-L2-H128 + attention	16	100.00	99.33	97.95

Table 1: Training, validation and test accuracy reported by the notebook for the explored configurations.

7 Training loss curves

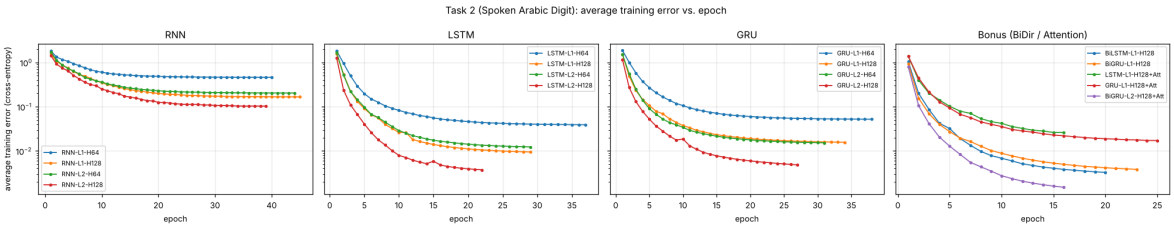


Figure 3: Average training error versus epoch for the explored Task 2 configurations.

8 Training and test accuracy comparison

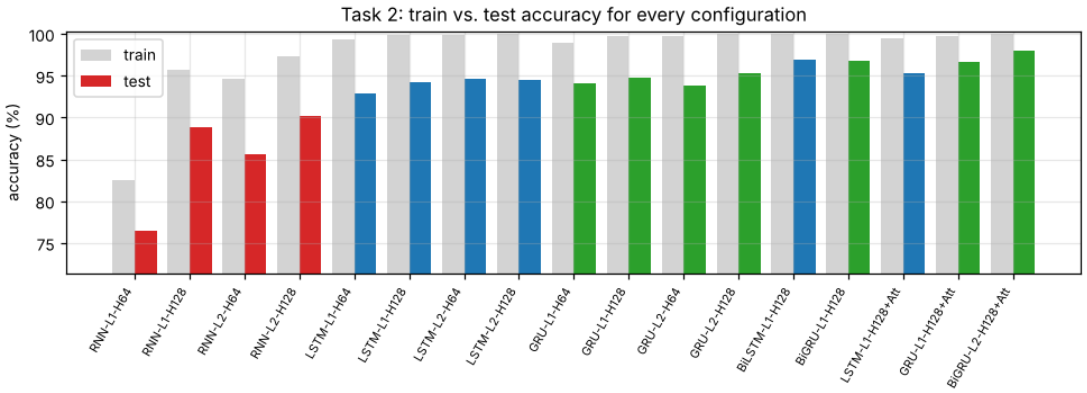


Figure 4: Training and test accuracy by model configuration.

The best plain recurrent models reported are RNN-L1-H128 (test 88.86%), LSTM-L2-H64 (test 94.64%) and GRU-L1-H64 (test 94.05%). The best overall configuration is BiGRU-L2-H128 with temporal attention, which reached 97.95% test accuracy.

9 Confusion matrices

A confusion matrix compares true labels (rows) with predicted labels (columns). High diagonal counts indicate correct predictions, while off-diagonal counts identify confusions between digit classes. Each test class contains 220 utterances in this dataset split.

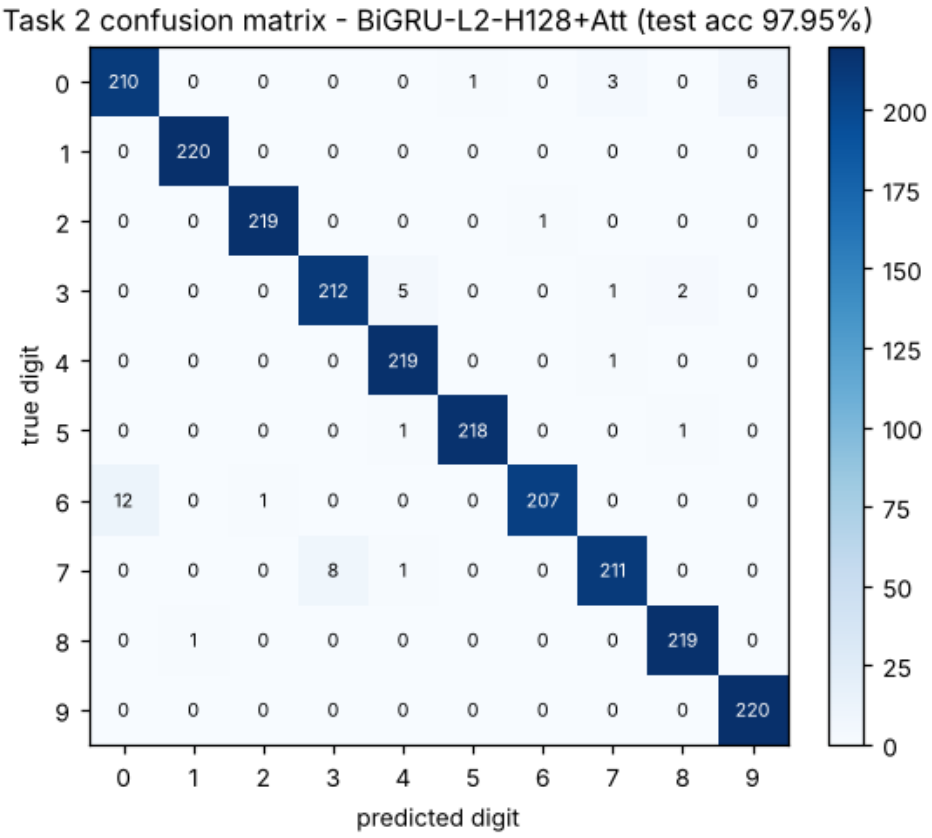


Figure 5: Confusion matrix for the plain RNN model (test accuracy 88.86%).

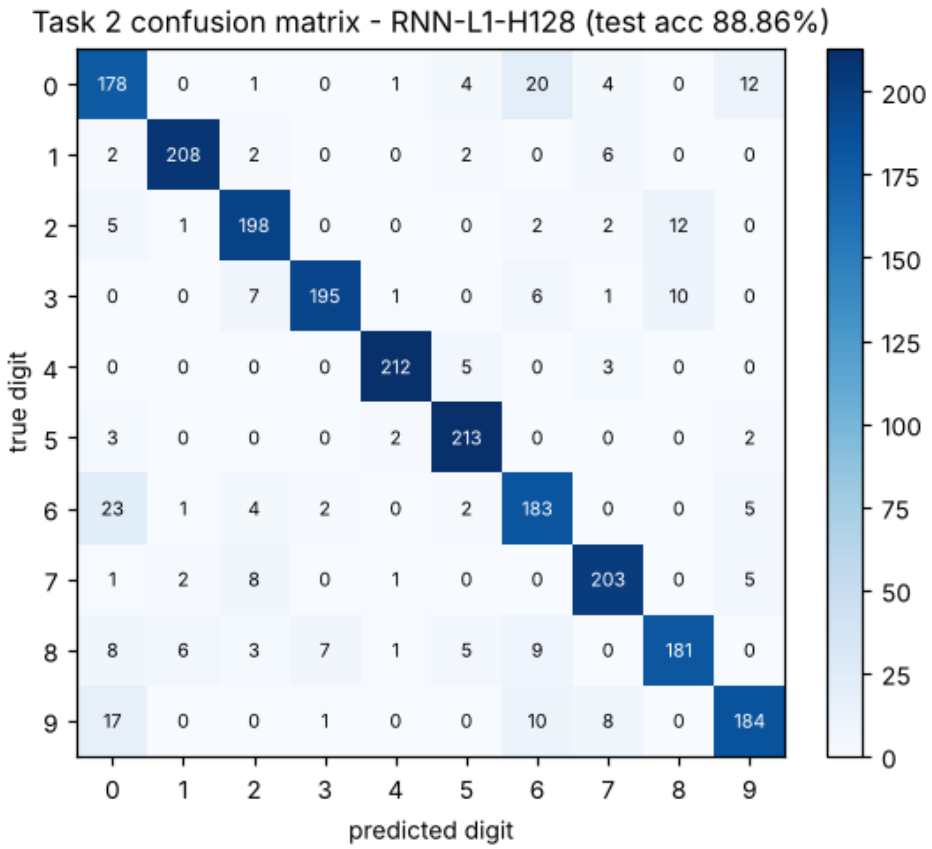


Figure 6: Confusion matrix for the plain LSTM model (test accuracy 94.64%).

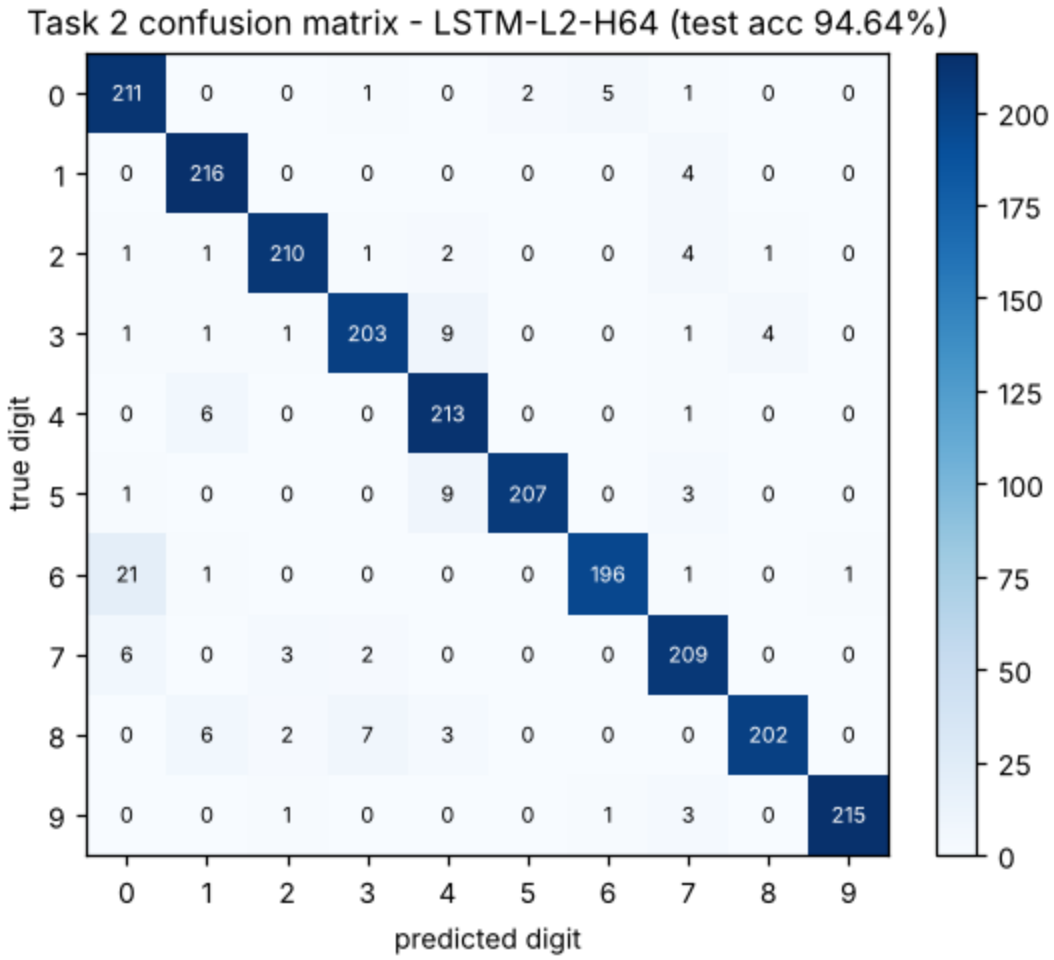


Figure 7: Confusion matrix for the best overall BiGRU + attention model (test accuracy 97.95%).

10 Gradient-flow analysis

Gradient-flow plots help explain how learning signals change across time steps, especially for long utterances. The notebook reports the ratio of earliest-frame to latest-frame gradient magnitude as follows: RNN, 1.014 × 10n¹n at initialization and 2.002 × 10n¹ after training; LSTM, 2.262 × 10nn at initialization and 8.384 × 10n¹ after training; GRU, 1.691 × 10nn at initialization and 3.417 × 10n¹ after training. These values suggest that gating can help preserve gradient information over long sequences, though the precise effect depends on training and model configuration.

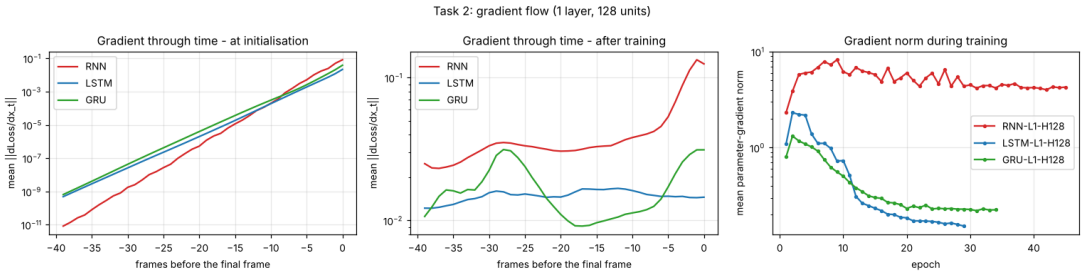


Figure 8: Gradient flow through time for recurrent models on long utterances.

11 Best-model detailed performance

The best overall model achieved 100.00% training accuracy, 99.33% validation accuracy and 97.95% test accuracy. Its classification report gives an overall test accuracy of 97.95% across 2,200 examples. Class-wise precision, recall and F1-score are high overall; the notebook output indicates comparatively lower recall for digit 6 (0.9409) and digit 0 (0.9545), suggesting these classes account for some remaining errors.

12 Conclusion

The experiments show that sequential MFCC features can be classified effectively with recurrent neural networks. LSTM and GRU models outperform the tested simple RNN configurations, and bidirectional processing with temporal attention provides the strongest result in this notebook. The best configuration, BiGRU-L2-H128 + attention, achieves 97.95% test accuracy. Future work may examine speaker robustness, confusion patterns, model size and inference time.